Faizan Abrar

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Summary

Python Backend Developer (Fresher) skilled in building REST APIs using FastAPI, PostgreSQL, Redis, Celery, and Docker. Developed a URL analytics platform that reduced median redirect latency by **50.5%** (**200ms to 99ms**) through Redis caching, and a deployed insurance prediction application using machine learning. B.Tech student in Electrical Engineering with strong foundations in DSA, OOP, and DBMS.

Skills

- **Languages:** Python, SQL, JavaScript, Java
- **Backend:** FastAPI, REST APIs, JWT Authentication, Pydantic, Celery, Redis
- **Databases:** PostgreSQL, SQLAlchemy (ORM)
- **Testing:** Pytest, Unit Testing, API Testing, Postman, Locust (Load Testing)
- **DevOps/Tools:** Docker, Git, Render (deployment), Redis CLI
- **Familiar:** React, HTML, CSS, Scikit-learn, SHAP, LLM APIs (Function Calling)
- **CS Fundamentals:** Data Structures & Algorithms, OOP, DBMS | 100+ LeetCode problems solved

Projects

SmartLink Analytics Platform (Full Stack) GitHub

- Developed a full-stack URL shortener with analytics using Python, FastAPI, React, PostgreSQL, Redis, and Celery.
- Improved URL redirect performance using Redis caching with a 1-hour TTL, reducing median latency by **50.5% (200ms to 99ms)** and increasing throughput by **28.6% (95.73 to 123.07 RPS)** under 50 concurrent users, with zero failures.
- Moved analytics updates (clicks, unique visitors, device usage, location, date-wise traffic) to Celery background workers to keep redirects fast; added a composite PostgreSQL index on (`url_id`, `timestamp`) for faster analytics queries.
- Added JWT authentication (signup, login, protected endpoints), rate limiting, and a paginated link history API.
- Integrated an LLM feature using function (tool) calling to generate traffic insights and engagement summaries.
- Wrote ~30 unit and API tests using Pytest on a separate test database.

Insurance Prediction System (Full Stack + ML) GitHub | Live | API

- Deployed (Docker, Render) a full-stack insurance prediction app using Python, FastAPI, PostgreSQL, and JavaScript, with authentication, predictions, and admin reporting.
- Designed REST APIs for registration/login, model inference, prediction storage, and paginated admin analytics, plus the PostgreSQL schema with SQLAlchemy ORM.
- Secured the app with Argon2, HTTP-only JWT cookies, role-based authorization, Pydantic validation, CSP headers, and rate limiting.
- Reduced model MAE from 4213 to 3941 (6.5% better) using log transformation and outlier analysis; added SHAP for explainable predictions.

Work Experience

Software Development Intern | *Scope For Change* May 2026 – June 2026

- Implemented **2 pages** (donation and volunteer registration) for the company website.
- Covered these pages with unit and integration tests and fixed bugs found before release.
- Worked with the design team on SEO and page speed improvements.

Education

Jamia Millia Islamia Expected 2027
B.Tech in Electrical Engineering New Delhi, India